Game Plan: Reaching Milestone 5 (25 Oct 2026)

Project: Where Naive Bayes Breaks (Group 2). Plan written 19 Oct 2026.

Reality check

M5 (A7 LLM camouflage, A8 poisoning, defences D1–D5) sits on top of everything from M1–M4: the NB classifier, data, the attack harness, and A1–A6. With until Sunday 25 Oct, this cannot be built serially by one person. First step today: message teammates and find out what already exists. Do not redo anything they have built.

If nobody has started, a suggested split:

- **Core (NB + features + harness):** one person, needed by Wednesday
- **Data (Fresh-100 labels, Enron, SpamAssassin loaders):** one person
- **Attacks A1–A6 and the analytic bound:** one person
- **You (suggested):** A7, A8, D1–D5, the M5 deliverables

If doing it all alone, follow the order below.

Day-by-day plan

Mon 19 Oct: Core classifier

- Set up the repo, venv, pytest, and a seed config.
- Write the tokenizer (re, lowercase) and load Enron-Spam with the temporal 70/30 split.
- Implement Multinomial NB (log-space, add-one smoothing) and Bernoulli NB.
- Pass the Example 13.1 and 13.2 tests. Do this first because everything else depends on it.

Tue 20 Oct: Evaluation harness

- P/R/F1 with spam as the positive class, bootstrap CIs, and a function that takes (model, documents, attack) and returns metrics.
- Expose weights w_t and per-document margin S(d), which attacks and defences all need.
- Get a clean baseline on Enron-Spam.

Wed 21 Oct: Minimum attack base and the slow job

- Implement A2 (passive good-word) and A3, so there is a working attack to test defences against.
- **Start A7 generation now.** It is the slowest step, so let it run in the background. Write the rewrite prompts (paraphrase only, then paraphrase plus benign padding), pick a local open-weight model or a free-tier API, and generate for about 300 spam messages. Defang URLs and replace contact details with placeholders.

Thu 22 Oct: Poisoning and RONI

- A8: inject {0.5, 1, 2, 5}% spam-labelled messages stuffed with ham vocabulary, retrain, and measure the damage.
- D5 (RONI-lite): reject a candidate training message if adding it drops held-out F1 by more than a threshold. Test it against A8.

Fri 23 Oct: Cheap defences

- D2: cap term frequency at c (c = 1 gives Bernoulli-like behaviour).
- D3: clip $|w_t| \leqslant \tau$.
- D4: segment-wise scoring (split into chunks, flag if any chunk is spam).
- Each is a few lines on top of the weights and scorer. Test each against A2/A3.

Sat 24 Oct: Encoding side

- A5 and A6 (minimal versions): hidden HTML text, and zero-width/homoglyph substitutions on the most spammy tokens.
- D1: unicodedata NFKC normalisation, strip zero-width and control characters, a small confusables map, and drop non-visible HTML text. Check that D1 recovers the A5/A6 damage.

Sun 25 Oct: Integrate

- Evaluate the A7 outputs against NB, with and without the defences.
- Produce before/after P/R/F1 tables and check the tests pass.
- Write up notes on what did not make it.

Triage if time runs short

Cut in this order (all can slip to Week 6 without blocking M5):

1. Reuters sanity check

2. MI and χ^2 feature selection (frequency selection is enough for now)
3. Full k × budget grid; run one or two settings instead
4. Multi-seed CIs; get single-seed results first

Do not cut the NB unit tests, the harness, A7, A8, or the defences themselves. Those are the M5 exit criteria.

Biggest risks

- **A7 getting stuck on LLM access or refusals.** Have the fallback ready: a rule-based paraphraser with templates.
- **Building before testing.** Get the Examples 13.1/13.2 tests green before anything else. A subtle smoothing bug will invalidate every later result.